1. **Configure a DNS server role on your Windows Server VM and create a new forward lookup zone named 'myapp.local' with an A record pointing to 192.168.1.50.**

- **1.Install DNS Server Role:Server Manager.**

Open Server Manager, click Add roles and features, select Role-based or feature-based installation, choose your server from the pool, check DNS Server under Server Roles, accept any required feature pop-ups, and complete the installation wizard.

Verification: DNS Server role shows as installed in Server Manager.

2.Open DNS Manager:Management Console.

Click on Tools in the top-right corner of Server Manager and select DNS to open the DNS Management console.

Verification: The DNS Manager console opens with your server listed in the left pane.

3.Create Forward Lookup Zone:myapp.local.

Expand your server node, right-click Forward Lookup Zones, select New Zone..., choose Primary zone, enter myapp.local as the Zone name, and click through the wizard using default settings until finished.

Verification: myapp.local appears under the Forward Lookup Zones folder.

4.Create A Record:192.168.1.50.

Right-click the newly created myapp.local zone and select New Host (A or AAAA).... Leave the Name field blank (to resolve myapp.local directly) or type a hostname, enter 192.168.1.50 in the IP address field, and click Add Host.

Verification: The A record appears in the zone's record list on the right pane.

2. **Set up the DHCP server role and create a new scope that assigns IP addresses from 192.168.1.100 to 192.168.1.120 with a lease duration of 8 hours.**

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1.Install DHCP Server Role: Server Manager.

Open Server Manager, click Add roles and features, select Role-based or feature-based installation, choose your server from the pool, check DHCP Server under Server Roles, accept any required feature pop-ups, and complete the installation wizard.

Verification: DHCP Server role shows as installed in Server Manager.

2.Complete DHCP Configuration: Post-Installation.
Click the notification flag (yellow warning icon) at the top of Server Manager and select Complete DHCP configuration to authorize the server in Active Directory and create security groups.

Verification: The configuration wizard completes successfully.

3.Create a New IPv4 Scope:DHCP Console.
Click on Tools in Server Manager and select DHCP. Expand your server hostname, right-click on IPv4, and select New Scope... to launch the wizard. Enter a name for the scope and click Next.

Verification: The New Scope Wizard opens.

4.Define IP Range and Subnet Mask:IP Address Range.
Enter the Start IP address as 192.168.1.100 and the End IP address as 192.168.1.120. Ensure the length and subnet mask (e.g., 24 / 255.255.255.0) match your network, then click Next.

Verification: The IP range is accepted by the wizard.

5.Configure Lease Duration to 8 Hours: Lease Duration.
Click through any exclusion or delay additions until you reach the Lease Duration page. Change the duration settings to 8 hours, click Next, select Yes, I want to configure these options now (to add Default Gateway/DNS if needed, or configure later), and click Finish.

Verification: The newly created scope appears under IPv4 and is marked as active with an 8-hour lease duration in its properties.

**3. Share a folder named 'MusicLibrary' on your Windows Server and set permissions so only users in the 'Students' group can read and write to it.

Hint: Use the 'Advanced Sharing' and 'Permissions' tabs to manage access.**

- **1.Create and Open Folder Properties:**Server local drive.
  Navigate to any drive (e.g., C: or D:) on your Windows Server, create a new folder named MusicLibrary, right-click it, select Properties, and go to the Sharing tab.
  Verification: The Sharing tab is open for the MusicLibrary folder.

- **2.Configure Advanced Sharing**:Share the folder.
  Click on the Advanced Sharing button, check the box for Share this folder, and then click on the Permissions button.
  Verification: The Advanced Share Permissions dialog box appears.

- **3.Restrict Access to Students Only:**Remove Everyone and Add Students.
  Select the Everyone group from the list and click Remove. Then click Add..., type Students, click Check Names, and press OK.
  Verification: Students appears in the group or user names list.

- **4.Assign Read and Change Permissions:**Share permissions.
  Select Students in the list and check Allow for both Read and Change (Write) permissions. Click Apply and OK on all open windows to save.
  Verification: The folder is successfully shared with Read and Change access restricted exclusively to the Students group.

4. Install the Print Server role, add a virtual printer (Microsoft XPS Document Writer), and share it as 'CampusPrinter' so network users can print to it.

- **1.Install the Print Server Role:**Server Manager.
  Open Server Manager, click on Add roles and features, select Role-based or feature-based installation, and choose your server from the pool. On the Server Roles screen, check Print and Document Services, click Add Features on the pop-up, click Next, and ensure Print Server is checked under Role Services. Proceed to complete the installation wizard.
  Verification: Print and Document Services role shows as installed in Server Manager.

2.Open Print Management: Print Management Console.
Click on Tools in the top-right corner of Server Manager and select Print Management to open the management console (printmanagement.msc).
Verification: The Print Management console window opens successfully.

3.Add a New Printer using Microsoft XPS: Add Printer Wizard.
In the left pane, expand Print Servers, click on your server name, and select Printers. Right-click in the middle pane and choose Add Printer... Select Install a new printer, choose Create a new port (Local Port), and click Next.
Verification: The port creation prompt appears.

4.Select Microsoft XPS Document Writer: Port and Driver Selection.
Choose Local Port as the port type, click Next, and enter a port name (e.g., XPSPort:) if prompted. On the driver selection screen, select Use an existing printer driver on the computer, choose Microsoft XPS Document Writer v4 from the list, and click Next.
Verification: The driver is selected without any missing dependency errors.

5.Configure 'CampusPrinter' Share Name: Name and Share Printer. Give the printer a recognizable name (e.g., Campus XPS Printer). On the sharing screen, select Share this printer and enter the share name explicitly as CampusPrinter. Click Next, finish the wizard, and close.
Verification: CampusPrinter appears in the Printers list inside the Print Management console with its shared status enabled.

5. Troubleshoot: A client PC is not receiving an IP address from your DHCP server. List 3 possible causes and describe how you would test and fix each one using Windows Server tools.

- **1. DHCP Scope Exhaustion (No IP Addresses Available)**
  - Cause: All available IP addresses within the defined scope range (e.g., 192.168.1.100 to 192.168.1.120) have already been leased out **to other** devices, leaving none for the new client.
  - **How to Test:**
    1. Open the DHCP console on your Windows Server (dhcpmgmt.msc).
    2. Expand your server, navigate to IPv4 -> Scope -> Address Pool.
    3. Check the visual allocation bar and summary to see if active leases equal the total pool size (100% utilization).

- **How to Fix:**
  - Expand the scope range (e.g., add more IP addresses) or shorten the lease duration so stale leases expire faster.
  - To shorten the lease duration, right-click your Scope -> Properties, and modify the Lease Duration to a shorter time.

2. DHCP Server Service Stopped or Unresponsive

- Cause: The DHCP Server service on Windows Server has stopped, crashed, or failed to start automatically after a reboot.
- **How to Test:**
  1. Open Services (services.msc) on the Windows Server.
  2. Scroll down and locate DHCP Server. Check its status column; if it does not say *Running*, the service is down.
- **How to Fix:**
  - Right-click DHCP Server and select Start (or Restart).
  - To prevent future occurrences, right-click the service, select Properties, and change the Startup type to Automatic.

3. Network Connectivity or DHCP Relay Agent (Router) Issue

- Cause: The client PC and the DHCP server are on different VLANs/subnets, and either physical network connectivity is broken, or a DHCP Relay Agent (IP Helper) has not been configured on the intervening router to forward DHCP broadcast requests across subnets.
- **How to Test:**
  - Run ipconfig /release followed by ipconfig /renew on the client PC. If it fails with an error stating "An error occurred while renewing interface... The DHCP client has obtained an IP address that is already in use on the network" or times out with an APIPA address (169.254.x.x), communication is blocked. ◦ Check the server's DHCP Event Viewer (eventvwr.msc -> *Applications and Services Logs -> Microsoft -> Windows -> DHCP-Server -> Operational*) to see if requests from that subnet are arriving.
- **How to Fix:**
  - Verify physical switch links and cable integrity between the client and server.
  - If subnets are separate, configure the IP Helper / DHCP Relay Agent address on your gateway router pointing to the Windows Server's static IP address (192.168.1.50).